

CENG381 Stochastic Processes

Classifying States - Question Set 1

Notation used throughout this question set:

$P(i, j)$	one-step probability of moving from state i to state j
$P^n(i, j)$	n -step probability: probability of being in state j after exactly n steps, starting from state i
Recurrent:	state i is recurrent if, starting from i , the chain returns to i with probability 1
Transient:	state i is transient if, starting from i , the chain has a positive probability of never returning to i
Absorbing:	state i is absorbing if $P(i, i) = 1$ (the chain never leaves)

Q1 (30 points). Consider the Markov chain with the following 5x5 transition matrix P (rows = current state, columns = next state):

	0	1	2	3	4
St 0	[0.0	1.0	0.0	0.0	0.0]
St 1	[0.5	0.0	0.5	0.0	0.0]
St 2	[0.0	0.5	0.0	0.5	0.0]
St 3	[0.0	0.0	0.0	0.4	0.6]
St 4	[0.0	0.0	0.0	0.0	1.0]

- Draw the transition diagram. Use arrows labelled with probabilities to show all transitions with positive probability.
 - Identify all communicating classes. (Two states i and j are in the same communicating class if state i can reach state j AND state j can reach state i , i.e. $P^n(i, j) > 0$ and $P^m(j, i) > 0$ for some $n, m \geq 1$.)
 - For each communicating class, state whether it is positive recurrent, null recurrent, or transient, and justify your answer.
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Q2 (35 points). A factory machine has four possible states: Running (R), Degraded (D), Under Repair (U), and Scrapped (S). Each day the machine transitions according to:

From R: stays R with prob 0.8, moves to D with prob 0.2

From D: stays D with prob 0.5, moves to U with prob 0.3, moves to S with prob 0.2

From U: returns to R with prob 0.7, stays U with prob 0.3

From S: stays S with prob 1.0 (absorbing)

- Write the 4x4 transition matrix P with rows and columns in the order (R, D, U, S).

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- b) Identify all communicating classes and classify each one as recurrent or transient.
- c) Starting from state R, what is the probability of eventually being absorbed into state S? Let $h(R)$, $h(D)$, $h(U)$ denote the probability of reaching S starting from R, D, U respectively. Set up and solve the system of linear equations for $h(R)$, $h(D)$, $h(U)$.
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Q3 (35 points). Consider the Markov chain on $\{0, 1, 2, 3, 4, 5\}$ with transition matrix P given by:

	0	1	2	3	4	5
St 0	[1/3	2/3	0.0	0.0	0.0	0.0]
St 1	[1/3	2/3	0.0	0.0	0.0	0.0]
St 2	[0.0	3/5	1/5	1/5	0.0	0.0]
St 3	[0.0	0.0	0.0	1/4	3/4	0.0]
St 4	[0.0	0.0	0.0	1/2	1/2	0.0]
St 5	[0.0	0.0	0.0	0.0	0.0	1.0]

- a) Draw the transition diagram and identify all communicating classes.
- b) Classify each communicating class as positive recurrent, null recurrent, or transient. Give a brief justification for each.
- c) Find the limiting (long-run) state probabilities $\pi(0)$, $\pi(1)$, ..., $\pi(5)$. For each transient class, the limiting probability of each state in that class is 0. For each positive recurrent class, solve $\pi * P_{\text{class}} = \pi$ restricted to that class, with probabilities summing to the weight of that class.